



Proposal: Azure DevOps Security Implementation Quote

For Leonardo Company - Canada

Prepared by Cloudstrucc Inc

Executive Summary

This quote outlines the scope and investment required to secure Azure DevOps for Leonardo Company Canada Inc. As a defense contractor with subsidiaries across multiple jurisdictions including the leonardo.com subscription / domain (leonardo.com) the organization requires a robust DevSecOps security framework that enables secure collaboration between developers working across international boundaries while protecting sensitive intellectual property and controlled goods.

Currently, development teams in Canada, Italy, and the United Kingdom share code and artifacts through various mechanisms including file share services and manual transfer methods. Leonardo Canada seeks to modernize this approach by adopting a structured Software Development Lifecycle (SDLC) and Application Lifecycle Management (ALM) methodology using Azure DevOps and Git-based source control. As code begins flowing through centralized repositories across national borders, a Zero Trust security architecture becomes essential—one that enforces strong identity controls, leverages managed identities for service authentication, and provides comprehensive audit capabilities. This security hardening initiative will establish the governance framework necessary to enable this transition: implementing managed identities and workload identity federation for secure authentication, configuring branch protection policies for controlled Git collaboration, deploying pipeline security controls for CI/CD processes, and enabling monitoring capabilities that provide visibility into cross-border development activities—all while protecting Leonardo's security interests and supporting productive international teamwork.

1. Business Context

1.1 Background

Leonardo Company Canada currently relies on file share services and other ad-hoc mechanisms to exchange code and development artifacts with the leonardo.com subscription/domain. To improve development efficiency, traceability, and security posture, Leonardo Canada is adopting Azure DevOps as its central DevSecOps platform, enabling a structured SDLC/ALM approach across the organization. This modernization initiative will establish:

- Centralized Git repositories replacing fragmented file-sharing methods for cross-border code collaboration
- CI/CD pipelines introducing automated, secure build and deployment processes for defense-related code
- Artifact management providing controlled versioning and distribution of software components
- Work item tracking enabling visibility and coordination across international development teams

1.2 Business Drivers

Driver	Description
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International Collaboration	Transition from file shares to secure, Git-based development workflows between Canada, Italy, and UK teams via the leonardo.com domain/subscription
Zero Trust Architecture	Eliminate implicit trust inherent in file-sharing; verify every access request regardless of origin
Data Sovereignty	Protect data flowing to and from Canada with appropriate controls as code moves to centralized repositories
Managed Identity Adoption	Remove stored credentials; use managed identities for all service authentication
Supply Chain Security	Protect against software supply chain attacks across the distributed development model
Audit Readiness	Maintain comprehensive audit trails for cross-border compliance reviews

1.3 Stakeholders

Stakeholder	Role	Interest
Leonardo Canada Development Teams	End Users	Secure collaboration workflows
leonardo.com Development Teams	End Users	Cross-tenant repository access
IT Security (All Regions)	Oversight	Policy enforcement, risk assessment
Compliance Office	Governance	Regulatory adherence, export controls

2. Security Requirements

2.1 Zero Trust Identity and Access Management

Req ID	Requirement	Priority	Rationale
IAM-001	All Azure DevOps access must use Microsoft Entra ID authentication	Critical	Centralized identity for all regions
IAM-002	Multi-factor authentication (MFA) must be enforced for all users	Critical	Defense-in-depth for international access
IAM-003	Conditional Access policies must enforce device compliance and location context	Critical	Zero Trust access verification
IAM-004	Managed identities must be used for all service-to-service authentication	Critical	Eliminate stored credentials

Req ID	Requirement	Priority	Rationale
IAM-005	Workload identity federation must replace service principal secrets	Critical	Secure cross-tenant pipeline authentication
IAM-006	Cross-tenant access policies must explicitly define allowed Leonardo subsidiaries	Critical	Control inter-company collaboration
IAM-007	Personal Access Tokens (PATs) must have maximum 90-day expiration	High	Reduce credential exposure window
IAM-008	Guest access from non-Leonardo tenants must be prohibited	Critical	Prevent unauthorized external access

2.2 Git Repository Security

Req ID	Requirement	Priority	Rationale
GIT-001	Branch policies must enforce pull request reviews for all shared repositories	Critical	Code review before merge
GIT-002	Production branches must require minimum 2 reviewers from appropriate regions	Critical	Separation of duties across teams
GIT-003	Secret scanning must be enabled on all repositories	Critical	Prevent credential leakage across borders
GIT-004	Push protection must block commits containing secrets	Critical	Real-time secret detection
GIT-005	Fork policies must restrict forking to within Leonardo organization	High	Prevent code exfiltration
GIT-006	Commit signing must be enforced for production branches	High	Code integrity and non-repudiation
GIT-007	Repository permissions must follow least privilege across all contributing teams	Critical	Minimize access scope

2.3 Pipeline and Build Security

Req ID	Requirement	Priority	Rationale
PIPE-001	YAML pipelines must be used exclusively (Classic pipelines deprecated)	High	Pipeline-as-code, version controlled
PIPE-002	Pipeline approval gates must be implemented for production deployments	Critical	Change control for releases

Req ID	Requirement	Priority	Rationale
PIPE-003	Service connections must use workload identity federation	Critical	No stored secrets for cross-tenant access
PIPE-004	Variable groups containing secrets must integrate with Azure Key Vault	Critical	Centralized secret management
PIPE-005	Self-hosted agents must be isolated and hardened	High	Secure build infrastructure
PIPE-006	Pipeline templates must enforce security scanning stages	High	Consistent security controls
PIPE-007	Marketplace extensions must be reviewed and approved before use	High	Supply chain security

2.4 Cross-Border Data Protection

Req ID	Requirement	Priority	Rationale
DATA-001	Audit logging must capture all cross-tenant access and activities	Critical	Visibility into international collaboration
DATA-002	Data residency for Canadian projects must remain in Canadian Azure regions	Critical	Data sovereignty compliance
DATA-003	Encryption in transit must use TLS 1.2+ for all Git operations	Critical	Protect data crossing borders
DATA-004	Artifact feeds must be private with explicit cross-tenant access grants	High	Control package distribution

2.5 Compliance and Audit

Req ID	Requirement	Priority	Rationale
AUD-001	Audit logs must be retained for minimum 7 years	Critical	Regulatory compliance
AUD-002	Security alerts must integrate with SOC tooling	High	Incident response
AUD-003	Access reviews must be conducted quarterly for cross-tenant permissions	High	Continuous compliance
AUD-004	Compliance reports must document cross-border data flows	Critical	Export control compliance

3. Zero Trust Security Framework

3.1 Managed Identity Strategy

Eliminating stored credentials is essential when development activities span multiple countries. The following managed identity approach ensures secure authentication without credential exposure:

Authentication Scenario	Solution
Pipeline to Azure Resources	System-assigned managed identity
Cross-tenant Pipeline Access	Workload identity federation
Service Connections	Managed identity with RBAC
Agent Authentication	Managed identity on VM/container
Key Vault Access	Managed identity with access policies

3.2 Cross-Tenant Collaboration Model

Allowed Leonardo Tenants:

Tenant	Direction	Use Case
leonardo.com	Bidirectional	Shared repository collaboration
All Others	Blocked	No external collaboration

Zero Trust Controls for Cross-Tenant Access:

- Explicit tenant allowlisting via Cross-Tenant Access Policies
- Conditional Access requiring compliant devices from partner tenants
- MFA enforcement for all cross-tenant authentication
- Session controls limiting token lifetime
- Continuous access evaluation for real-time policy enforcement

3.3 ITSG-33 Alignment

ITSG-33 Control	Azure DevOps Implementation
AC-2 Account Management	Entra ID integration, access reviews
AC-3 Access Enforcement	RBAC, repository permissions
AC-4 Information Flow	Cross-tenant policies, data residency
AC-6 Least Privilege	Managed identities, minimal permissions
AU-2 Audit Events	Audit streaming, activity logs
IA-2 Identification and Authentication	MFA, Conditional Access
IA-8 Identification (Non-Org Users)	Cross-tenant access policies

ITSG-33 Control	Azure DevOps Implementation
SC-7 Boundary Protection	Network restrictions, private endpoints
SC-8 Transmission Confidentiality	TLS encryption

4. Project Plan

4.1 Timeline Overview

Target Completion: March 31, 2026

Phase 1: Assessment & Planning	Jan 6 – Jan 12
Phase 2: Zero Trust Identity	Jan 12 – Jan 24
Phase 3: Cross-Tenant Configuration	Jan 27 – Feb 7
Phase 4: Git Repository Security	Feb 10 – Feb 21
Phase 5: Pipeline Security	Feb 24 – Mar 7
Phase 6: Monitoring & Compliance	Mar 10 – Mar 21
Phase 7: Documentation & Handover	Mar 24 – Mar 31

4.2 Detailed Work Breakdown

Phase 1: Assessment & Planning (Jan 6 - Jan 12)

Task	Deliverable
Current state assessment	Security assessment report
Cross-tenant access inventory	Collaboration mapping
Gap analysis	Gap analysis document
Remediation planning	Approved implementation plan

Phase 2: Zero Trust Identity Configuration (Jan 12 - Jan 24)

Task	Deliverable
Configure Entra ID-only authentication	Configuration documentation
Implement Conditional Access policies	CA policy set
Deploy managed identities for service connections	Managed identity configurations
Configure workload identity federation	Federation trust relationships
PAT policy enforcement	Token governance

Phase 3: Cross-Tenant Configuration (Jan 27 - Feb 7)

Task	Deliverable
Cross-tenant access policy design	Policy framework
leonardo.com tenant configuration	collaboration enabled
Cross-tenant Conditional Access	Partner tenant controls
Testing and validation	Test results

Phase 4: Git Repository Security (Feb 10 - Feb 21)

Task	Deliverable
Branch policy implementation	Branch policies deployed
Secret scanning enablement	Scanning active
Push protection configuration	Protection rules
Repository permission review	Least privilege applied
Commit signing policy	Signing requirements

Phase 5: Pipeline Security (Feb 24 - Mar 7)

Task	Deliverable
Secure pipeline template development	Template library
Environment approval workflows	Approval gates
Key Vault integration	Secret management
Extension governance	Approved extension list

Phase 6: Monitoring & Compliance (Mar 10 - Mar 21)

Task	Deliverable
Audit streaming configuration	Log Analytics integration
Security monitoring dashboard	Azure Workbook
Alert rules for cross-tenant activity	Alert policies
Compliance reporting	Report templates

Phase 7: Documentation & Handover (Mar 24 - Mar 31)

Task	Deliverable
Build book finalization	Complete build book
SOP documentation	Operating procedures

Task	Deliverable
Knowledge transfer sessions	Training delivered
Project closeout	Final acceptance

4.3 Milestones

Milestone	Target Date	Success Criteria
M1: Assessment Complete	January 12, 2026	Approved implementation plan
M2: Zero Trust Identity Active	January 24, 2026	Managed identities deployed
M3: Cross-Tenant Configured	February 7, 2026	leonardo.com enabled
M4: Repositories Secured	February 21, 2026	Branch policies active
M5: Pipelines Hardened	March 7, 2026	Secure templates deployed
M6: Monitoring Operational	March 21, 2026	Dashboards live
M7: Project Complete	March 31, 2026	Documentation delivered

5. Investment Summary

5.1 Professional Services

Resource	Description
Frederick Pearson, Lead Architect	Security architecture, Zero Trust design, managed identity implementation, cross-tenant configuration, and technical leadership
Support Engineer	Implementation support, testing, documentation, and knowledge transfer

Professional Services Total: \$22,500

5.2 Infrastructure and Licensing

Item	Description	Cost
Implementation	Initial setup, configuration, and deployment activities	\$1,000
Azure DevOps Licensing	Client responsibility - existing or new licenses	Client Cost
Azure Log Analytics	Client responsibility - ingestion and retention	Client Cost
Azure Key Vault	Client responsibility - Premium tier for managed identities	Client Cost
Microsoft Entra ID P2	Client responsibility - Conditional Access, PIM	Client Cost

Infrastructure Total (Cloudstrucc): \$1,000

5.3 Training and Knowledge Transfer

Item	Description	Cost
Knowledge Transfer Sessions	Two half-day sessions covering security controls, managed identity operations, and cross-tenant management	\$2,500
Documentation Package	Build book, SOPs, runbooks, and quick reference guides	\$1,000

Training Total: \$3,500

5.4 Quote Summary

Category	Cost (CAD)
Professional Services	\$22,500
Infrastructure (Cloudstrucc)	\$1,000
Training & Knowledge Transfer	\$3,500
Subtotal	\$27,000
HST (13%)	\$3,510
Total Investment	\$30,510

6. Post-Implementation Support

Cloudstrucc Inc. offers ongoing support options following project completion:

6.1 Support Rates

Resource	Hourly Rate (CAD)
Frederick Pearson, Lead Architect	\$125/hour
Support Engineer	\$90/hour

6.2 Support Services Available

- Security configuration adjustments
- New managed identity deployments
- Cross-tenant access modifications
- Incident response assistance
- Quarterly access reviews
- Compliance audit support
- Pipeline security enhancements

Support engagements are available on an as-needed basis with a minimum 2-hour engagement.

7. Risk Considerations

Risk	Likelihood	Impact	Mitigation
Cross-tenant configuration complexity	Medium	High	Phased rollout, thorough testing
Resistance to managed identity adoption	Medium	Medium	Clear documentation, training
Pipeline disruption during migration	Medium	High	Non-production testing, rollback plans
Coordination across Leonardo subsidiaries	Medium	Medium	Early stakeholder engagement
Timeline dependencies on partner tenants	Medium	Medium	Parallel workstreams where possible

8. Assumptions and Dependencies

8.1 Assumptions

- Leonardo Canada has Azure DevOps Organization Administrator access
- Leonardo.com will participate in cross-tenant configuration
- Microsoft Entra ID P2 licensing is available or will be procured
- Azure subscription is available for Log Analytics and Key Vault
- Client will provide timely access and approvals

8.2 Dependencies

- Cross-tenant access requires coordination with leonardo.com IT teams
- Conditional Access policies may require Entra ID P2 activation
- Managed identity deployment requires Azure resource permissions

